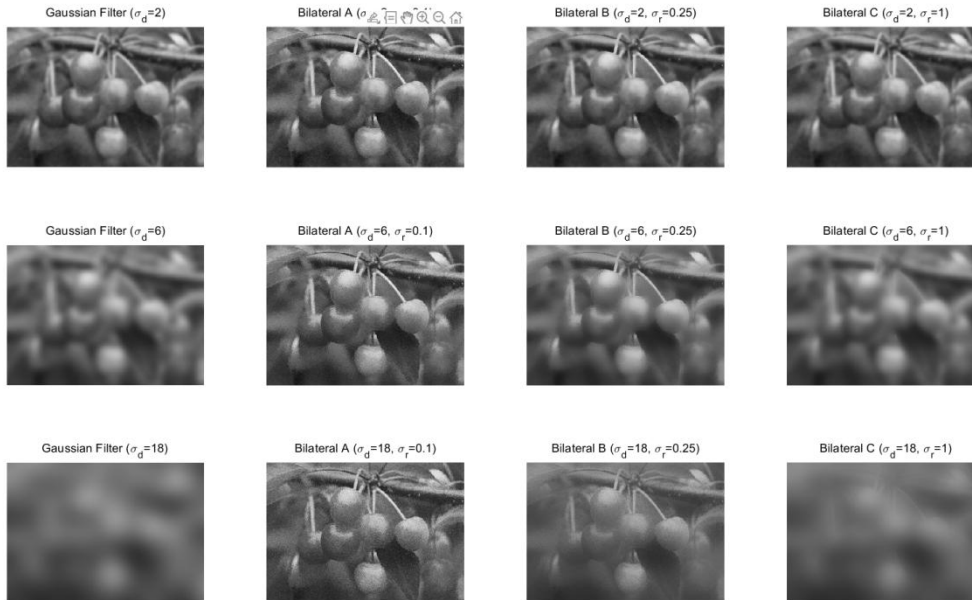


Experiment #1

Image Smoothing with Bilateral Filtering



The experimental results are shown above. The experiment compares the performance of the Gaussian Filter and the Bilateral Filter under 12 different parameter configurations.

According to the results, we can know that how parameters σ_d and σ_r influence the behavior of the bilateral filter and differences between the Gaussian filter and the bilateral filter.

The parameter σ_d controls the size of the neighborhood in terms of spatial distance. It determines how far the filter looks for neighboring pixels. Increasing σ_d increases the spatial extent of the filter and decreasing σ_d limits the filter to a small local area.

The parameter σ_r controls how much the filter penalizes differences in intensity. It determines what constitutes an edge. Increasing σ_r makes the filter more tolerant of intensity differences and decreasing σ_r enforces strict edge preservation.

Gaussian filter smooths the image based only on spatial distance while bilateral filter is based on both spatial distance and intensity difference. Gaussian filter blurs edges but bilateral filter preserves shape edges. As σ_r approaches infinity, the bilateral filter loses its sensitivity to intensity differences and behaves exactly like a standard Gaussian filter.